

Environmental Science Assignment

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Course Name : Environmental Science

Date : 21-06-2026
Year / Term : I Year Term IV
Assessment : Assignment - 1

1. Explain the structure and functions of an ecosystem.

Structure of an Ecosystem

An **ecosystem** is a community of living organisms interacting with each other and with their physical environment.

The structure of an ecosystem consists of two main components:

A. Biotic Components (Living Organisms)

1. **Producers (Autotrophs)** – Green plants and algae that prepare food through photosynthesis.
2. **Consumers (Heterotrophs)** – Organisms that depend on others for food.
 - Primary consumers (herbivores)
 - Secondary consumers (carnivores)
 - Tertiary consumers (top carnivores)
3. **Decomposers** – Bacteria and fungi that break down dead organic matter and recycle nutrients.

B. Abiotic Components (Non-Living Factors)

- Sunlight
- Air
- Water
- Soil
- Temperature
- Minerals and nutrients

Functions of an Ecosystem

1. **Productivity** – Production of organic matter by plants.
2. **Energy Flow** – Transfer of energy from producers to consumers.
3. **Nutrient Cycling** – Recycling of nutrients through biogeochemical cycles.
4. **Decomposition** – Breakdown of dead organisms and waste materials.
5. **Ecological Regulation** – Maintains environmental balance and supports life.

2. What is ecological balance? Explain its importance.

Ecological Balance

Ecological balance is the state in which living organisms and their environment exist in harmony, maintaining stable interactions and sustainable use of resources.

Importance of Ecological Balance

1. Maintains biodiversity.
2. Ensures proper functioning of food chains and food webs.
3. Regulates climate and environmental conditions.
4. Supports nutrient recycling.
5. Prevents overpopulation or extinction of species.
6. Provides clean air, water, and fertile soil.
7. Ensures long-term survival of ecosystems and human beings.

Example

In a forest ecosystem, plants, herbivores, carnivores, and decomposers interact in a balanced way. Disturbance in one group affects the entire ecosystem.

3. Explain food chain with a suitable example.

Food Chain

A **food chain** is a sequence of organisms through which energy and nutrients pass from one organism to another by feeding.

Characteristics

- Shows a single pathway of energy flow.
- Begins with producers.
- Ends with decomposers.

Example

Grass → Grasshopper → Frog → Snake → Eagle

Explanation:

- Grass is the producer.
- Grasshopper is the primary consumer.
- Frog is the secondary consumer.
- Snake is the tertiary consumer.
- Eagle is the top consumer.

Importance

- Demonstrates energy transfer.
- Maintains ecosystem stability.
- Helps understand interdependence among organisms.

4. What are trophic levels? Explain different trophic levels.

Trophic Levels

A **trophic level** is the position occupied by an organism in a food chain based on its feeding relationship.

Different Trophic Levels

1st Trophic Level – Producers

- Green plants and algae.
- Convert solar energy into chemical energy.

2nd Trophic Level – Primary Consumers

- Herbivores that feed on plants.
- Examples: Deer, Rabbit, Grasshopper.

3rd Trophic Level – Secondary Consumers

- Carnivores that feed on herbivores.
- Examples: Frog, Lizard.

4th Trophic Level – Tertiary Consumers

- Carnivores feeding on secondary consumers.
- Examples: Snake.

5th Trophic Level – Top Consumers

- Apex predators.
- Examples: Eagle, Tiger.

Decomposers

- Act on all trophic levels.
- Examples: Bacteria and fungi.

5. Explain ecological pyramids and their types.

Ecological Pyramid

An **ecological pyramid** is a graphical representation showing the relationship among organisms at different trophic levels in terms of number, biomass, or energy.

Types of Ecological Pyramids

A. Pyramid of Numbers

Shows the number of organisms at each trophic level.

Example: Grassland Ecosystem

- Many grasses
- Fewer rabbits
- Even fewer foxes

Shape: Usually upright.

B. Pyramid of Biomass

Shows the total dry weight (biomass) of organisms at each trophic level.

Shape:

- Upright in terrestrial ecosystems.
- Inverted in some aquatic ecosystems.

C. Pyramid of Energy

Shows the amount of energy available at each trophic level.

Shape: Always upright.

Importance

- Illustrates feeding relationships.
- Shows energy transfer and productivity.
- Helps study ecosystem functioning.

6. Explain the pyramid of energy and its importance.

Pyramid of Energy

The **pyramid of energy** represents the amount of energy available at different trophic levels of a food chain.

10% Law

According to the **10% Law** proposed by Raymond Lindeman, only about **10% of the energy** is transferred from one trophic level to the next, while the remaining 90% is lost as heat and metabolic activities.

Characteristics

1. Always upright.
2. Energy decreases at each successive trophic level.
3. Cannot be inverted.
4. Represents actual energy flow in an ecosystem.

Importance

1. Explains energy transfer in ecosystems.
2. Helps understand ecosystem productivity.
3. Shows why food chains are usually short.
4. Demonstrates energy loss at each trophic level.
5. Useful in environmental and ecological studies.